Deep Learning Practical Oral Questions and Answers

1. Linear Regression using Deep Neural Network (Boston Housing Dataset)

Q1. What is Linear Regression?

Answer: A supervised learning algorithm used for predicting continuous values.

Q2. What is Deep Neural Network?

Answer: A neural network having multiple hidden layers.

Q3. What is Boston Housing Dataset?

Answer: A dataset used for predicting house prices based on different features.

Q4. What is supervised learning?

Answer: Learning using labeled input and output data.

Q5. What is regression?

Answer: Predicting continuous numerical values.

Q6. What is a feature?

Answer: An input variable used for prediction.

Q7. What is target variable in Boston dataset?

Answer: House price.

Q8. What is an epoch?

Answer: One complete iteration over the training dataset.

Q9. What is batch size?

Answer: Number of samples processed before updating weights.

Q10. What is activation function?

Answer: Function applied to neuron output.

Q11. What activation is used in regression output layer?

Answer: Linear activation.

Q12. What is loss function for regression?

Answer: Mean Squared Error (MSE).

Q13. What is optimizer?

Answer: Algorithm used to update weights.

Q14. What is Adam optimizer?

Answer: Adaptive optimization algorithm used in deep learning.

Q15. What is overfitting?

Answer: Model performs well on training data but poorly on testing data.

Q16. What is underfitting?

Answer: Model fails to learn training data properly.

Q17. What is normalization?

Answer: Scaling data into smaller range.

Q18. Why preprocessing is important?

Answer: To improve model performance.

Q19. What is training data?

Answer: Data used to train the model.

Q20. What is testing data?

Answer: Data used to evaluate model performance.

2. Classification using Deep Neural Network

Q21. What is classification?

Answer: Predicting categorical outputs.

Q22. What is binary classification?

Answer: Classification into two classes.

Q23. What is multiclass classification?

Answer: Classification into more than two classes.

Q24. What is OCR?

Answer: Optical Character Recognition.

Q25. What is OCR letter recognition dataset?

Answer: Dataset used for recognizing alphabets.

Q26. What is IMDB dataset?

Answer: Movie review dataset used for sentiment analysis.

Q27. What is sentiment analysis?

Answer: Finding positive or negative opinion from text.

Q28. What is label encoding?

Answer: Converting labels into numeric values.

Q29. What is one-hot encoding?

Answer: Representing categories as binary vectors.

Q30. Which activation function is used in hidden layers?

Answer: ReLU.

Q31. Which activation function is used in binary classification output?

Answer: Sigmoid.

Q32. Which activation function is used in multiclass classification output?

Answer: Softmax.

Q33. What is sigmoid function?

Answer: Activation producing output between 0 and 1.

Q34. What is softmax function?

Answer: Converts outputs into probabilities.

Q35. What is cross entropy loss?

Answer: Loss function used in classification.

Q36. What is categorical cross entropy?

Answer: Loss function for multiclass classification.

Q37. What is binary cross entropy?

Answer: Loss function for binary classification.

Q38. What is accuracy?

Answer: Percentage of correct predictions.

Q39. What is confusion matrix?

Answer: Table showing prediction results.

Q40. What are true positives?

Answer: Correctly predicted positive samples.

Q41. What are false positives?

Answer: Incorrectly predicted positive samples.

Q42. What are false negatives?

Answer: Incorrectly predicted negative samples.

Q43. What is precision?

Answer: Ratio of correct positive predictions.

Q44. What is recall?

Answer: Ability to find all positive samples.

Q45. What is F1-score?

Answer: Harmonic mean of precision and recall.

3. Convolutional Neural Network (CNN)

Q46. What is CNN?

Answer: Deep learning model mainly used for image processing.

Q47. What is convolution?

Answer: Operation used to extract image features.

Q48. What is kernel/filter?

Answer: Small matrix used for feature extraction.

Q49. What is feature map?

Answer: Output produced after convolution.

Q50. What is pooling?

Answer: Reducing image dimensions.

Q51. What is max pooling?

Answer: Selecting maximum value from region.

Q52. What is flattening?

Answer: Converting matrix into vector.

Q53. What is dropout?

Answer: Technique used to reduce overfitting.

Q54. What is image classification?

Answer: Categorizing images into classes.

Q55. What is MNIST Fashion Dataset?

Answer: Dataset containing fashion clothing images.

Q56. What is plant disease detection?

Answer: Identifying plant diseases using images.

Q57. Why CNN is suitable for images?

Answer: Automatically extracts spatial features.

Q58. What is image preprocessing?

Answer: Resizing and normalizing images.

Q59. What is data augmentation?

Answer: Generating modified training images.

Q60. What are epochs in CNN?

Answer: Iterations during training.

Q61. What is stride?

Answer: Step size during convolution.

Q62. What is padding?

Answer: Adding extra pixels around image borders.

Q63. What is grayscale image?

Answer: Image having shades of gray only.

Q64. What is RGB image?

Answer: Image containing red, green, and blue channels.

Q65. Applications of CNN?

Answer: Medical imaging, face recognition, autonomous vehicles.

4. Recurrent Neural Network (RNN) – Google Stock Price Prediction

Q66. What is RNN?

Answer: Neural network designed for sequential data.

Q67. What is time series data?

Answer: Data collected over time intervals.

Q68. Why RNN is used for stock prediction?

Answer: RNN remembers previous information.

Q69. What is sequence data?

Answer: Ordered data points.

Q70. What is hidden state in RNN?

Answer: Memory of previous inputs.

Q71. What is Google stock dataset?

Answer: Historical stock prices of Google.

Q72. What is stock price prediction?

Answer: Forecasting future stock prices.

Q73. What is LSTM?

Answer: Long Short-Term Memory network.

Q74. Why LSTM is better than basic RNN?

Answer: It solves vanishing gradient problem.

Q75. What is vanishing gradient problem?

Answer: Gradients become too small during training.

Q76. What is sequential dependency?

Answer: Current output depends on previous data.

Q77. What is forecasting?

Answer: Predicting future values.

Q78. What is timestep?

Answer: Single unit in sequence.

Q79. What is sliding window?

Answer: Using previous values to predict next value.

Q80. What optimizer is commonly used in RNN?

Answer: Adam optimizer.

Q81. What loss function is used in stock prediction?

Answer: Mean Squared Error.

Q82. What is normalization in stock prediction?

Answer: Scaling stock values.

Q83. What is training sequence?

Answer: Input sequence used for learning.

Q84. Applications of RNN?

Answer: Speech recognition, NLP, stock prediction.

Q85. Difference between CNN and RNN?

Answer: CNN processes images while RNN processes sequences.

Advanced Viva Questions

Q86. What is deep learning?

Answer: Subset of machine learning using neural networks.

Q87. What is machine learning?

Answer: Systems learning patterns from data.

Q88. Difference between AI, ML and DL?

Answer: DL is subset of ML and ML is subset of AI.

Q89. What is TensorFlow?

Answer: Open-source deep learning framework by Google.

Q90. What is Keras?

Answer: High-level API for deep learning.

Q91. What is PyTorch?

Answer: Deep learning framework developed by Meta.

Q92. What is GPU?

Answer: Processor used for parallel computations.

Q93. Why GPU is used in deep learning?

Answer: Faster matrix calculations.

Q94. What is backpropagation?

Answer: Technique for updating neural network weights.

Q95. What are weights in neural networks?

Answer: Parameters adjusted during training.

Q96. What are biases?

Answer: Additional parameters improving learning.

Q97. What is gradient descent?

Answer: Optimization algorithm minimizing loss.

Q98. What is learning rate?

Answer: Controls weight update size.

Q99. What is hyperparameter?

Answer: Parameter set before training.

Q100. What is model evaluation?

Answer: Checking model performance.

Q101. What is train-test split?

Answer: Dividing dataset for training and testing.

Q102. What is validation data?

Answer: Data used during training evaluation.

Q103. What is transfer learning?

Answer: Using pre-trained models.

Q104. What is early stopping?

Answer: Stopping training to avoid overfitting.

Q105. Applications of Deep Learning?

Answer: Healthcare, NLP, finance, robotics, autonomous vehicles.

Total Questions Covered: 105